

Student Dashboard and Academic Management System

1. Abstract

The Student Dashboard & Academic Management System is a web-based platform designed to provide students with a centralized place to organize and monitor academic activities. The application combines student registration and login, profile access, dashboard summaries, assignment management, task tracking, study scheduling, course/subject management, attendance monitoring, deadline tracking, and basic progress visualization. The frontend is developed using React.js and styled using Tailwind CSS. Node.js and Express.js provide the backend REST APIs, while MongoDB stores application data. JWT is used for authentication. Postman can be used to test APIs, and Git/GitHub provide source-code version control and repository management.

2. Introduction

Students often manage assignments, study schedules, subjects, tasks, and deadlines using separate applications. This project provides a centralized academic management platform where these activities can be organized through a single dashboard. The system is designed around a personalized student experience. After authentication, the student can access the dashboard and use modules for courses, assignments, attendance, tasks, study planning, profile management and logout. It helps students organize assignments, tasks, subjects, study schedules, deadlines, and academic progress efficiently. The system provides an easy-to-use and responsive dashboard that makes academic management more organized and convenient.

3. Problem Statement

A student needs a simple and organized system to track academic responsibilities, deadlines, study sessions, and course-related progress without depending on multiple disconnected tools.

4.Objectives

- Secure student registration and login.
- Provide a personalized student dashboard.
- Manage assignments and academic tasks.
- Organize study schedules and subjects.
- Track deadlines and academic progress.
- Provide a responsive and user-friendly interface.
- Maintain data through MongoDB and expose backend functionality through REST APIs.

5.Features and Modules

1.Authentication: Registration, login, JWT-based authentication, protected access, and logout

2. Student Dashboards: Centralized overview of courses, assignments, attendance, tasks, recent activity, and progress.

3. Assignment Managements: Create and manage assignments, due dates, subjects, priorities, and completion status.

4. Task Management: Create tasks, set priorities/due information, and mark tasks as completed.

5. Study Planner: Create study sessions with subject/topic, date, time, priority, and completion status.

6. Course/Subject Management: View enrolled courses/subjects, instructors, course codes, and progress.

- 7. Attendance:** Show overall attendance and course-wise attendance percentages.
- 8. Deadline Tracking:** Display upcoming and pending academic deadlines.
- 9. Responsive UI:** Provide a clean interface suitable for desktop and smaller screens.
- 10. Progress Tracking:** Display basic course and activity progress using percentages and progress bars.

6.Technology Stack

- 1. React.js & Tailwind CSS – Frontend and UI design.
- 2. Node.js & Express.js – Backend and API development.
- 3. MongoDB – Database management.
- 4. JWT, Postman & GitHub – Authentication, API testing, and version control.

7.Submission Deliverables

Deliverable	Evidence / Contents
1. Authentication Module	Create Account and Login screenshots
2. Student Dashboard	Student Dashboard screenshot
3. Assignment Management System	Assignments screenshot
4. Study Planner Module	Study Planner screenshot
5. Subject Management Module	My Courses screenshot
6. Database Design	MongoDB collections/schema documentation
7. API Documentation	REST endpoint documentation and Postman testing
8. GitHub Repository	Complete source code, README, .gitignore, and version history

8.Screenshot and Module Demonstration

8.1.Authentication

The Create Account screen collects student details such as full name, email, password, student ID, department, and semester. The Login screen allows an existing student to authenticate.

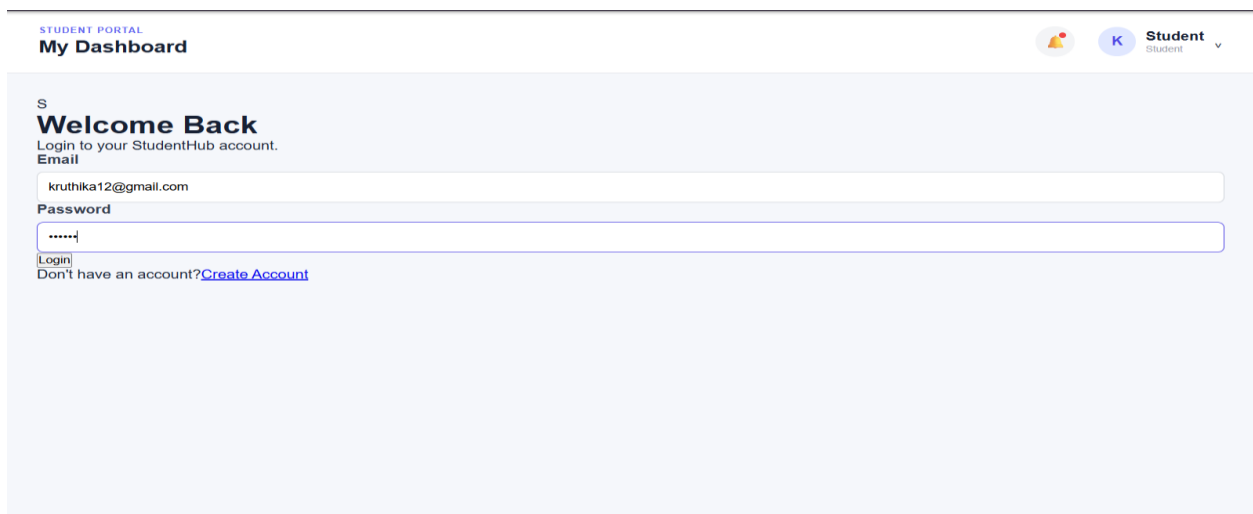
a. Create Account / Student Registration



The screenshot shows the 'Create Account' page of the Student Portal. The header includes 'STUDENT PORTAL' and 'My Dashboard'. On the right, there are notification and profile icons, with the profile labeled 'Student'. The main content area is titled 'Create Account' with the subtitle 'Join StudentHub and manage your academic life.' Below this are several input fields: 'Full Name' (filled with 'KRUTHIKA H K'), 'Email' (filled with 'kruthika12@gmail.com'), 'Password' (filled with '*****'), 'Student ID' (filled with 'ST001'), 'Department' (filled with 'AIML'), and 'Semester' (a dropdown menu filled with 'Semester 7'). At the bottom, there are links for 'Create Account' and 'Already have an account? Login'.

The registration interface demonstrates the Student Registration requirement.

b. Student login

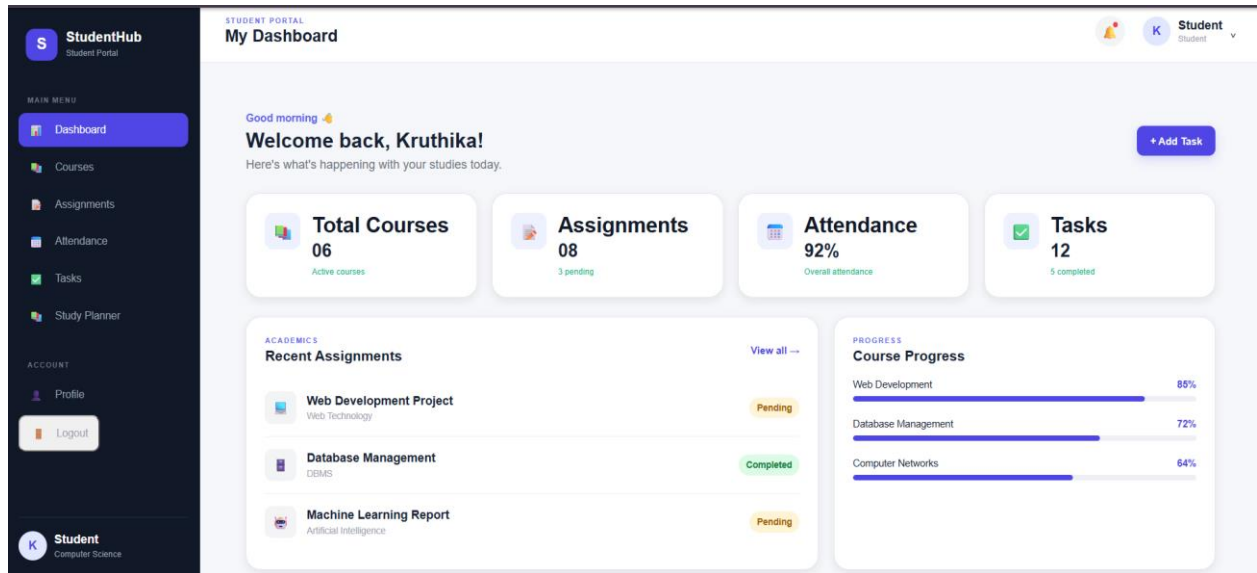


The screenshot shows the 'Welcome Back' login page of the Student Portal. The header is identical to the registration page. The main content area is titled 'Welcome Back' with the subtitle 'Login to your StudentHub account.' Below this are two input fields: 'Email' (filled with 'kruthika12@gmail.com') and 'Password' (filled with '*****'). At the bottom, there are links for 'Login' and 'Don't have an account? Create Account'.

The login interface demonstrates the Login requirement. Logout is available from the application sidebar.

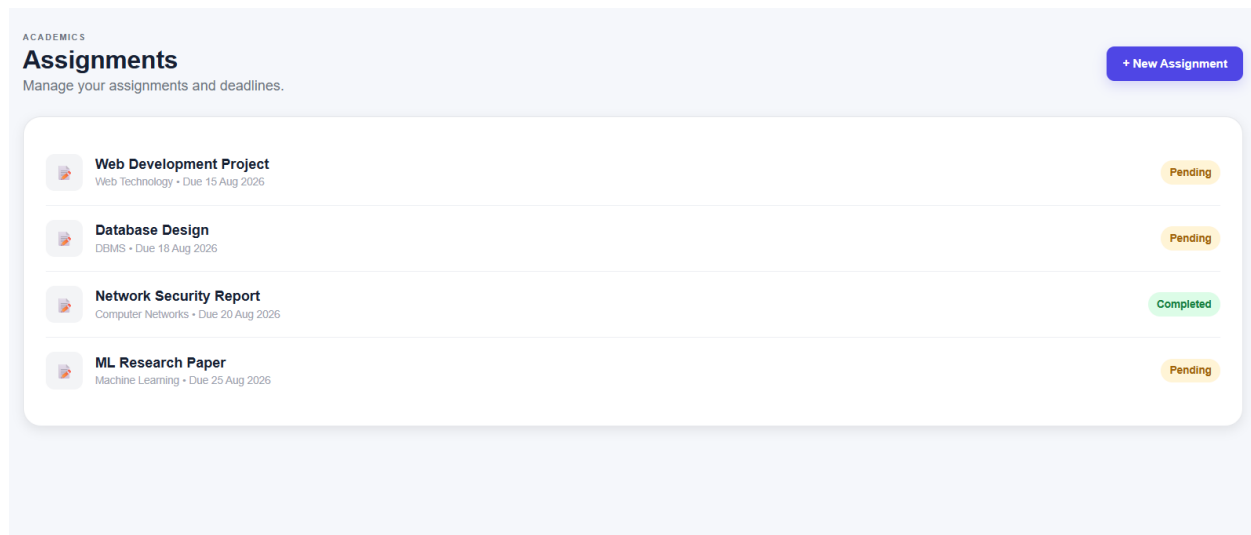
8.2.Student Dashboard

The dashboard acts as centralized academic overview. It presents total courses, assignments, attendance, tasks, recent assignments, and course progress in one screen.



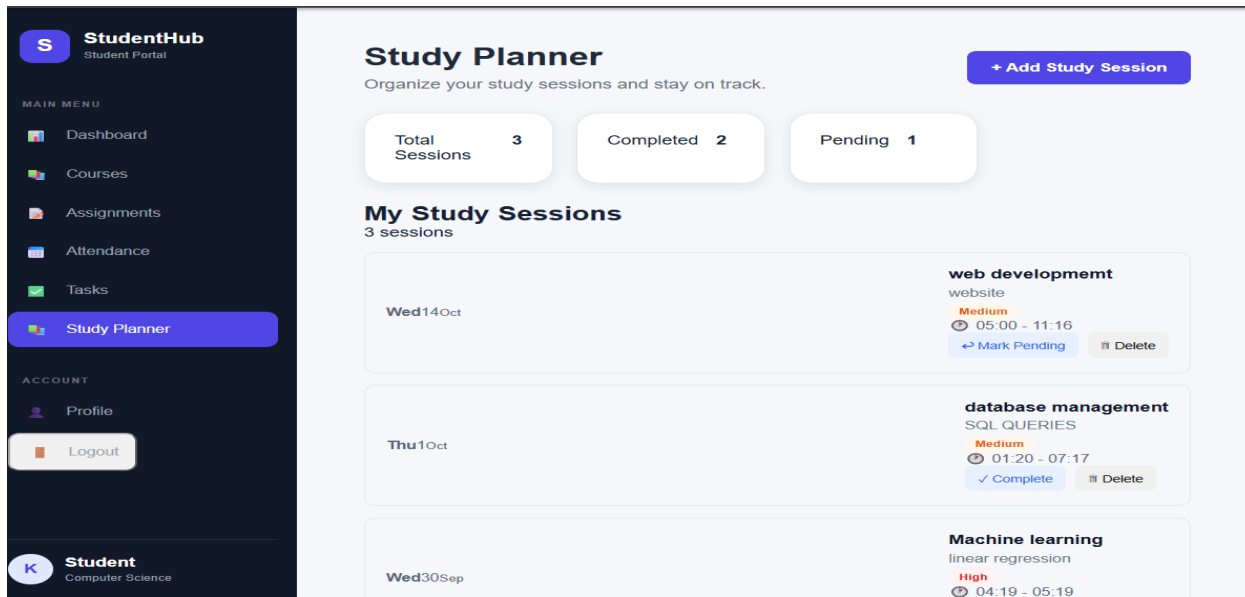
8.3. Assignment Management

It displays assignment titles, associated subjects, due dates, and status such as Pending or Completed



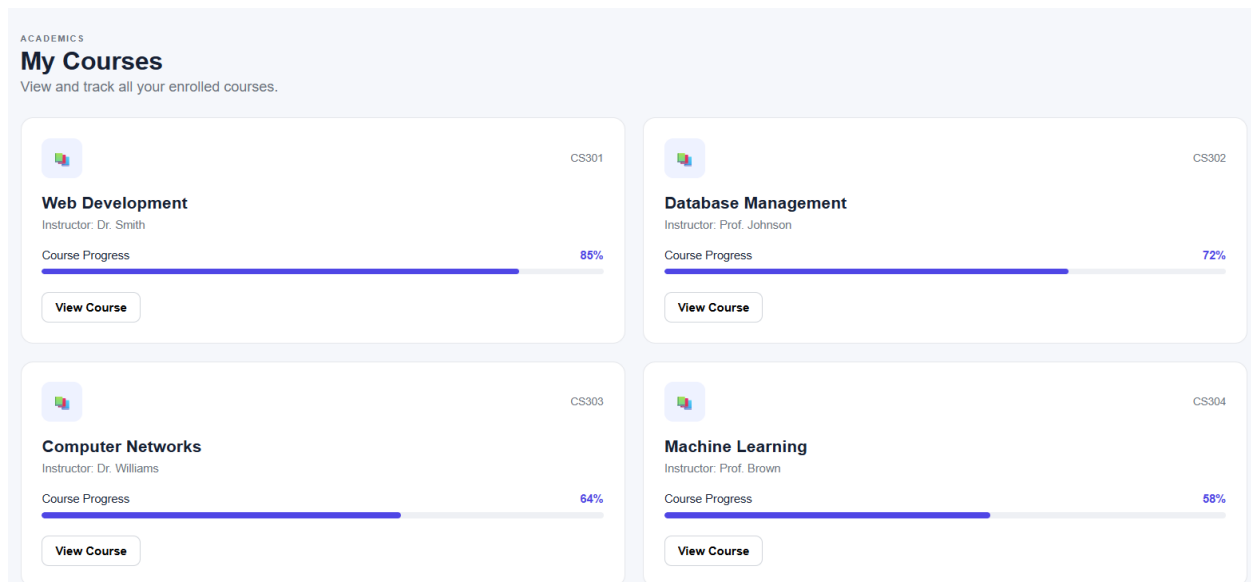
8.4.Study Planner Module

The Study Planner organizes study sessions and provides summary counts for total, completed, and pending sessions. Individual sessions show date, topic, priority, time, and status controls.



8.5. Subject Management Module

The Courses page displays enrolled subjects with course codes, instructor information, and course progress. This supports course/subject management and progress tracking.



8.6. Task Creation and Tracking

The Tasks page supports academic task tracking. Tasks can be shown as pending or completed and can carry priority indicators.

PRODUCTIVITY

My Tasks

Organize and track your academic tasks.

+ Add Task

☐ Complete React project
Due today

High

☐ Prepare DBMS presentation
Due tomorrow

Medium

☒ Submit network assignment
Completed

Done

8.7. Attendance and Progress Tracking

The Attendance page shows an overall attendance percentage and course-wise attendance. This provides a simple academic monitoring feature and contributes to the progress-oriented dashboard.

ACADEMICS

Attendance

Track your attendance for all courses.

92%
Overall

Excellent attendance!
Your attendance is above the required minimum.

Web Development	95%
Database Management	90%
Computer Networks	88%
Machine Learning	94%

9.Database Design

MongoDB is used as the database layer. The data can be separated into collections so that each student can have personal academic records.

Collection	Suggested Fields
Users	name, email, password Hash, student Id, department, semester, created At
Subjects	name, code, instructor, progress, user Id
Assignments	title, description, subject Id, due Date, priority, status, user Id
Tasks	title, due Date, priority, completed, userId
Study Schedules	Subject Id, topic, date, start Time, end Time, priority, status, user Id
Attendance	Subject Id, attended Classes, total Classes, percentage, user Id

10.API Documentation

The backend exposes REST APIs for authentication and academic modules. Postman can be used to send requests, inspect responses, and verify authentication and CRUD operations.

Authentication

Method	Endpoint	Purpose
POST	/api/auth/register	Register a student
POST	/api/auth/login	Authenticate a student
GET	/api/auth/profile	Retrieve authenticated profile
POST	/api/auth/logout	Logout / invalidate client session

Assignments

Method	Endpoint	Purpose
POST	/api/assignments	Create assignment
GET	/api/assignments	List assignments
PUT	/api/assignments/:id	Update assignment
DELETE	/api/assignments/:id	Delete assignment

Subjects

Method	Endpoint	Purpose
POST	/api/subjects	Create subject
GET	/api/subjects	List subjects
PUT	/api/subjects/:id	Update subject
DELETE	/api/subjects/:id	Delete subject

Study Planner

Method	Endpoint	Purpose
POST	/api/study	Create study session
GET	/api/study	List study sessions
PUT	/api/study/:id	Update study session
DELETE	/api/study/:id	Delete study session

11. Testing

Testing should verify both frontend behavior and backend API behavior.

Test Case	Action	Expected Result
Registration	Enter valid student details and submit	Account is created and validation is shown for invalid input
Login	Enter registered email and password	Student is authenticated and redirected to dashboard
Logout	Select Logout from sidebar	User session/token is cleared and login screen is shown
Assignment	Add, update, complete, and delete an assignment	Assignment list and status update correctly
Task	Create a task and mark it complete	Task status changes to completed
Study Planner	Create a study session and change its status	Session appears with correct details/status
Subject	View/add/update subjects	Course/subject information is displayed correctly
Attendance	Open attendance page	Overall and course-wise percentages are displayed
Responsive UI	Open app on different screen sizes	Layout remains usable and readable

12. Future Scope

- Email or push notifications for upcoming deadlines.
- Calendar integration for study schedules.
- File upload for assignment submissions.
- Attendance entry and automated percentage calculation.

13.Expected Result

- Students can register and log in to the StudentHub application.
- Authenticated users can access a personalized academic dashboard.
- Assignments and deadlines can be organized and monitored.
- Academic tasks can be created and marked as completed.
- Study sessions can be planned and tracked.
- Courses/subjects can be displayed with progress information.
- Attendance can be viewed overall and by subject.
- The interface provides clear navigation through dashboard, courses, assignments, attendance, tasks, study planner, profile, and logout

14.Conclusion

The Student Dashboard & Academic Management System provides a centralized platform for managing important student activities. The implemented screens demonstrate the core user-facing deliverables: authentication, dashboard, assignments, study planning, courses/subjects, tasks, and attendance. The project also defines the required backend, database, API, and GitHub deliverables. With the remaining backend integration, API testing evidence, database screenshots/schema, and GitHub repository link attached, the report can serve as a complete submission document.